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INTRODUCTION

The rapid growth of digital streaming platforms has transformed the way audiences consume entertainment content worldwide. Among these platforms, Amazon Prime Video has emerged as a major player, offering a vast and diverse library of movies and television shows across multiple genres, languages, and regions. With the increasing volume and variety of content, analysing such datasets has become essential for understanding trends, user preferences, and content distribution strategies.

This project focuses on performing content analysis using a dataset derived from Amazon Prime Video, leveraging the capabilities of MongoDB. Unlike traditional relational databases, MongoDB provides a flexible, document-oriented data model that is well-suited for handling semi-structured and unstructured data. This makes it particularly effective for managing streaming platform datasets, which often include nested attributes such as cast, genres, ratings, and production details.

The primary objective of this project is to explore and analyse various aspects of the dataset, including the distribution of movies and TV shows, genre popularity, release trends over time, and regional content availability. Additionally, the project aims to examine patterns related to ratings, durations, and contributions of actors and directors. By utilizing MongoDB's aggregation framework, meaningful insights can be extracted efficiently from large and complex datasets.

Through this analysis, the project seeks to demonstrate how NoSQL databases can be effectively used for real-world data exploration and decision-making. The findings may provide valuable insights into content production trends and audience engagement, highlighting the importance of data-driven strategies in the competitive streaming industry.

DATASET DESCRIPTION

The dataset used in this project is based on content available on Amazon Prime Video and consists of detailed information about movies and television shows hosted on the platform. It is a structured yet flexible dataset that contains multiple attributes describing each title, making it suitable for analysis using a NoSQL database like MongoDB.

Each record in the dataset represents a single title, which may be either a movie or a TV show. The dataset includes a variety of fields such as title name, content type (Movie/TV Show), director, cast, country of origin, date added to the platform, release year, rating (such as age classification), duration, genre (listed as multiple categories), and a brief description. These attributes provide both categorical and textual data, enabling a wide range of analytical operations.

One of the key characteristics of this dataset is the presence of multi-valued and nested fields. For example, the “cast” and “genre” fields often contain multiple entries within a single record, making them ideal for storage in array formats supported by MongoDB. Additionally, some records may contain missing or inconsistent values, reflecting real-world data challenges such as incomplete metadata or variations in formatting. This requires preprocessing steps such as data cleaning, normalization, and transformation before performing meaningful analysis.

The dataset also spans multiple years and includes content from various countries, allowing for temporal and regional analysis. This makes it possible to study trends such as the growth of content over time, the dominance of certain genres, and the contribution of different regions to the platform’s library. Furthermore, the inclusion of descriptive text fields enables potential use of text analysis techniques for deeper insights.

Overall, the dataset provides a comprehensive view of streaming content and serves as a strong foundation for exploring patterns, trends, and relationships within entertainment data. Its compatibility with MongoDB’s document-oriented structure allows efficient querying and aggregation, making it highly suitable for this content analysis project.

IMPLEMENTATION

The implementation of this project is carried out using MongoDB to store, manage, and analyse the dataset obtained from Amazon Prime Video. The process begins with data preparation, where the dataset is cleaned and transformed into a JSON format suitable for MongoDB.

Once the dataset is prepared, it is imported into MongoDB as a collection, where each document represents a single title (movie or TV show). The flexible schema of MongoDB allows attributes such as genres, cast, and countries to be stored as arrays, enabling efficient querying of multi-valued fields. Indexing is applied on commonly queried fields such as type, release year, and genre to improve query performance.

The core of the implementation involves using MongoDB's aggregation framework to perform data analysis. Various aggregation pipelines are designed to extract meaningful insights from the dataset. Operations such as `$match` are used to filter specific subsets of data, while `$group` is applied to categorize and compute statistics like counts and averages. The `$unwind` operator is used to decompose array fields such as genres and cast, allowing detailed analysis at an individual element level. Additionally, `$sort` and `$project` are used to organize and shape the output data as required.

Several analytical queries are executed to explore key aspects of the dataset. These include identifying the distribution of movies and TV shows, determining the most popular genres, analysing content trends over different years, and examining contributions from various countries. Further analysis is conducted to evaluate ratings and durations, as well as to identify frequently appearing actors and directors.

The results of these queries are then interpreted and can be visualized using charts or graphs for better understanding. Tools such as MongoDB Compass or external visualization platforms may be used to present the findings effectively. Overall, the implementation demonstrates how MongoDB can be utilized to efficiently handle large-scale, semi-structured datasets and perform complex analytical operations, making it a suitable choice for content analysis in streaming platforms.

SAMPLE OUTPUTS

1. Average Movie duration:

The analysis provides the average duration of the type “Movie” from the Amazon Prime Dataset. Through this we can able to get which movie is of larger duration than other.

```
db.getCollection('prime').aggregate([
  { $match: { type: 'Movie' } },
  { $sort: { duration: -1 } },
  {
    $addFields: {
      duration_num: {
        $arrayElemAt: [
          { $split: ['$duration', ' '] },
          0
        ]
      }
    }
  }
])
```

2. Content by Rating:

This aggregation analysis includes the output of all the content present in the dataset whether it is a Movie or TV Show, it will display all the ratings given to all the content.

```
db.getCollection('prime').aggregate([
  {
    $group: {
      _id: '$rating',
      count: { $sum: 1 }
    }
  },
  { $sort: { _id: 1 } },
  { $limit: 10 }
])
```

3. Genre Distribution by Content Type:

This output demonstrates the distribution of genre type of each individual content by type that is, Movie or TV Shows. This provides a brief description of how and what kind of genre a content belongs to.

```
db.getCollection('prime').aggregate([
  {
    $addFields: {
      genres: { $split: ['$listed_in', ','] }
    }
  },
  { $unwind: { path: '$genres' } },
  {
    $group: {
      _id: { type: '$type', genre: '$genres' }
    }
  }
])
```

4. Year Wise Content Growth:

This analysis displays how the content whether it is movie or TV Shows, how was the overall growth of the content as per year wise representation.

```
db.getCollection('prime').aggregate([
  {
    $group: {
      _id: '$release_year',
      count: { $sum: 1 }
    }
  },
  { $sort: { _id: -1 } }
],
{ maxTimeMS: 60000, allowDiskUse: true })
```

5. Content Release per Year:

This output shows the distribution of the content by release year of the content and also displays the total number of releases per year for that particular year.

```
db.getCollection('prime').aggregate([
  {
    $group: {
      _id: 'release_year',
      count: { $sum: 1 }
    }
  },
  { maxTimeMS: 60000, allowDiskUse: true }
]);
```

6. Number of Movies VS T.V Shows:

This demonstrates the total number of Movies and TV Shows present in the whole dataset which provides the overall head count of the total content present in the Amazon Prime dataset.

```
db.getCollection('prime').aggregate([
  {
    $group: {
      _id: {
        year: '$release_year',
        type: '$type'
      },
      count: { $sum: 1 }
    }
  },
]);
```

7. Top 10 Directors based on Content:

The analysis focuses on the top 10 directors present in the dataset and contributed the most in terms of number of Movies/TV Shows released by their name and work.

```
db.getCollection('prime').aggregate([
  { $match: { director: { $ne: null } } },
  {
    $group: {
      _id: '$director',
      count: { $sum: 1 }
    }
  },
  { $sort: { count: -1 } },
  { $limit: 10 }
```

8. Top 10 Countries producing Content:

This demonstrates and displays the top 10 countries from the dataset who made the greatest number of contents out of all the countries involved in the dataset. This signifies the total contribution by that particular country.

```
db.getCollection('prime').aggregate([
  {
    $group: {
      _id: '$country',
      count: { $sum: 1 }
    }
  },
  { $sort: { count: -1 } },
  { $limit: 10 }
],
```


DATA VISUALIZATION

This is the overall content visualization of all the aggregation analysis we performed in the MongoDB. This also demonstrates clean and clear visuals of the analysis along with particular representation of the charts used for the visualization.

